

DAG-GFlowNet vs Gibby

1 Bayesian Structure Learning

Definition 1. *Bayesian Network* (G, θ_G) is a probabilistic model over d random variables $\mathbf{X} = \{X_1, \dots, X_d\}$, whose joint distribution factorizes according to DAG G :

$$p(\mathbf{X}|G, \theta_G) = \prod_{i=1}^d p(X_i | Pa_G(X_i), \theta_i), \quad (1)$$

where $Pa_G(X_i)$ is the set of parents of the node X_i , θ_i parameters that define the probability distribution.

We are given a dataset $\mathcal{D} = (\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)})$ of N observations of random variable $\mathbf{X}$. The problem is to characterize the posterior distribution $P(G|D)$ over Bayesian Network structures that model these observations. Using Bayes formula, we can derive:

$$p(G|\mathcal{D}) = \frac{p(G, D)}{p(D)} = \frac{\int_{\theta_G} p(G, D, \theta_G) d\theta_G}{p(D)} = \frac{\int_{\theta_G} p(D|G, \theta_G) p(\theta_G|G) p(G) d\theta_G}{p(D)}. \quad (2)$$

Since we do not have access to $p(D)$, for any DAG G , we define its reward as the joint probability density function that equals to $p(G|D)$ up to constant $P(D)$:

$$R(G) = p(G, D) = \int_{\theta_G} p(D|G, \theta_G) p(\theta_G|G) p(G) d\theta_G. \quad (3)$$

Then, making assumptions of independence for parameter priors and of modularity for both parameter and DAG priors, we can derive closed form expression for the reward $R(G)$ as a product of local scores that only depend on individual variables and their parents in G :

$$R(G) = \prod_{i=1}^d \text{LocalScore}(X_i | Pa_G(X_i)). \quad (4)$$

Thus, we can calculate the reward for every DAG G . However, we still can not calculate the normalizing constant for the posterior since the number of possible DAGs is very large, making it computationally impossible. What we can do is to try to sample from the posterior $P(G|D)$.

To evaluate the quality of sampling method we will use *MAD*, i.e., *the maximum absolute deviation*:

$$\text{MAD} = \max_{i,j} |p_{ij} - \hat{p}_{ij}|, \quad (5)$$

where $\hat{p}_{ij}$ is the relative frequency of arc ij in the sampled DAGs and p_{ij} is the respective exact probability we compute by an exponential-time algorithm.

2 Experimental setup

For now, we assume each X_k to be a categorical random variable, each $p(X_k|\text{Pa}_G(X_k), \theta_G)$ to be parametrized by fully conditional probabilistic tables, and assign the parameters independent Dirichlet priors. Then, the local score from (4) is equal to BDeu score, parametrized by single constant called *equivalent sample size*.

First, we are going to compare *DAG-GFlowNet* with *Gibby* on *Asia* and *Sachs* datasets.